

SAI DHRUV YELLANKI

College Park, Maryland (Ready to Relocate)

☎ 240-960-1300 ✉ dhruvsai@umd.edu 🔗 [linkedin.com/in/saidhruvyellanki](https://www.linkedin.com/in/saidhruvyellanki) 🐙 github.com/dhruvyellanki19 📁 [Portfolio](#)

Professional Summary

AI Engineer and M.S. in Data Science student with 3+ years of experience fine-tuning LLMs (LoRA, QLoRA, RLHF), building Retrieval-Augmented Generation (RAG) systems, and developing multi-agent workflows for intelligent automation, predictive analytics, time-series forecasting, and anomaly detection. Proficient in Python, PySpark, Azure Databricks, and Kubernetes, with domain expertise in healthcare, including GDPR, HIPAA, and FHIR compliant solutions that enhance clinical workflows and medical decision support.

Education

University of Maryland

Masters of Science in Data Science; GPA: 3.89/4.0

August 2024 – May 2026

College Park, MD, USA

Experience

Aurelius Tech & Talent Solutions

AI Engineer Intern

May 2020 – August 2020

Ellicott City, MD, USA

- Developed a proposal generation system using a RAG pipeline with Azure OpenAI embeddings, PostgreSQL pgvector, and Redis caching, reducing healthcare document retrieval time by 65% and achieving 500 ms query responses.
- Fine-tuned the GPT-OSS-20B model to orchestrate a network of agents build using GPT-4 for retrieval, drafting, validation, and compliance analysis, achieving 79% precision and 74% recall in healthcare compliance detection aligned with HIPAA standards.
- Deployed the system through containerized LLMops workflows using Docker, Kubernetes, and Azure DevOps CI/CD, automating deployment, scaling, and monitoring to maintain 95% uptime and reduce release cycles by 40%.
- Optimized system performance through distributed processing and caching, increasing document throughput by 35% and enabling fast, regulation-compliant healthcare proposal generation.

Watzmann Consulting

Data Analyst – Predictive Analytics & Machine Learning

June 2023 – June 2024

Hyderabad, TG, India

- Engineered GDPR- and FHIR-compliant pipelines in Azure Databricks using PySpark and SQL to integrate patient demographics, clinical observations, and surgical schedules, improving data consistency and supporting analytics for over 200K+ patient records.
- Designed a stacked-ensemble Length-of-Stay prediction model using XGBoost, LightGBM, and Bi-LSTM on temporal and clinical data, achieving a Mean Absolute error less than 0.6 days and reducing average patient stay by 10% through discharge planning.
- Created an operating room scheduling system using Bi-LSTM models trained on surgical and workload data, combined with mixed-integer linear programming optimization to forecast procedure durations and reduce operating room idle time by 18%.
- Implemented MLOps pipelines with MLflow and Azure Machine Learning for model tracking, versioning, and deployment, and built Power BI dashboards to monitor accuracy and resource utilization, improving overall surgical department efficiency by 15%.

Ramphal Technologies

Data Science Intern

December 2022 – May 2023

Hyderabad, TG, India

- Trained system-performance prediction models using PyTorch and XGBoost to forecast application latency and server load, improving prediction accuracy by 12% and reducing average request delay by 7.5 ms.
- Constructed a real-time event pipeline with RabbitMQ and PostgreSQL to process 10K+ telemetry events per second, reducing false alerts and cutting reporting latency by 30% through Tableau dashboards and improved operational visibility for engineering teams.

Projects

Drug Repurposing and Biomedical Discovery | *Python, LangChain, Neo4j, LangGraph, Docker, ChromaDB, LLMs* | [GitHub Link](#)

- Built a drug-repurposing system using BioBERT and SciSpaCy for biomedical text extraction, Neo4j for knowledge graph modeling, and a RAG pipeline with LangChain and ChromaDB, mapping 50K+ drug-gene-disease links with 88% retrieval accuracy.
- Orchestrated multi agent workflows using LangGraph,LLMs (Llama3, Mistral7B) to automate reasoning and generate explainable, evidence-based insights, improving research speed 3x and reducing manual analysis 60% while maintaining 84% factual accuracy.
- Deployed the platform via Streamlit and Docker, implementing CI/CD pipelines (GitHub Actions) for automated testing and versioned updates, enabling 40% faster releases and ensuring reliable, production-ready performance.

AI Medical Chatbot for Automated Healthcare Triage | *Python, LoRA Fine-Tuning, RAG, Milvus, vLLM,Docker* | [GitHub Link](#)

- Developed an AI medical chatbot using Llama3 and Mixtral-7B, LangChain, and Milvus, and fine-tuned models with LoRA on 2K+ medical dialogues; evaluated with ROUGE and retrieval metrics, improving response precision by 30%.
- Deployed a multimodal RAG pipeline using Docker, Gradio, and vLLM to optimize inference and automate preliminary medical triage, reducing routine staff workload by 40% and improving telemedicine response efficiency.

Technical Skills

Programming Languages & Databases: Python (Pandas, NumPy, Scikit-learn, PySpark, TensorFlow, PyTorch, SpaCy, NLTK, Matplotlib, Seaborn, Plotly, Dash), SQL, R,PostgreSQL, Redis, MySQL, MongoDB.

Tools & Technologies: Docker, Kubernetes, Azure (Machine Learning, DevOps, Databricks), MLflow, FastAPI, vLLM,Git, Power BI, Tableau, Grafana, CI/CD Pipelines.

Machine Learning & Analytics: Predictive Modeling, Deep Learning, Time Series Forecasting (Prophet, ARIMA), MLOps, model stacking, Model Deployment & Monitoring, Anomaly Detection, hyperparameter tuning, cross-validation.

AI & LLM Engineering: NLP (NER, summarization, sentiment analysis, embeddings, transformers), fine-tuning(LoRA,QLoRA, RLHF), RAG, multi-agent orchestration (LangChain, LlamaIndex, LangGraph, AutoGen), Prompt Engineering, inference (vLLM).